

Input Validation

Team name: Project Team 4 (four)

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Purpose: This document lists all the places where a user can type or select information in the TP1 app. It explains the rules for each input field. The app should not accept input that is blank, wrong, unsafe, or too long. All input must be checked before the app uses it or saves it. When input is wrong/incorrect, the app should show a clear error message that tells the user how to fix it.

General Input Validation Rules: The following rules apply to all text input fields in the application:

1. Required fields cannot be blank.
2. Leading and trailing spaces should be removed before validation.
3. All text fields must have a maximum length.
4. Input must be validated before it is processed.
5. Invalid input must not be saved to the database.
6. Error messages must clearly explain what the user needs to fix.
7. Extremely long input must be rejected to prevent crashes or security weaknesses.

A major security concern is that attackers may paste very long strings into input fields to see if the application crashes. To reduce this risk, every text field has a reasonable maximum length.

Admin Home Page Input Fields: The Admin Home Page allows the admin to manage users, send invitations, set one-time passwords, update accounts, delete users, list users, and add or remove roles.

Screen Output	Field	Requirement	Validation Rule	Error Message
Admin Home Page	Email Address	Yes	Must be a valid email address and must not exceed XXX characters.	Enter a valid email address, such as projectteam4@gmail.com
Admin Home Page	Role	Yes	Must be one of the approved roles: Admin, Student, Instructor, or Staff.	Please select a valid role.
Admin Home Page	One-Time Password	Yes	Must follow password rules and must not exceed XX characters.	One-time password does not meet password requirements.

Login Page Input Fields:

Screen	Field	Required?	Validation Rule	Error Message
Login Page	Username	Yes	Must be X-XX characters. May contain letters, numbers, periods, hyphens, and underscores. Cannot contain spaces.	Username must be X-XX characters and may only contain letters, numbers, periods, hyphens, and underscores.
Login Page	Password	Yes	Cannot be blank. Must not exceed XX characters.	Password is required and must not exceed XX characters.

Username Validation Rules: The username validation rules are based on the team's HW1 username validation work.

A valid username must follow these rules:

- Username is required.
- Username must be between X and XX characters.
- Username may contain uppercase letters.
- Username may contain lowercase letters.
- Username may contain numbers.
- Username may contain periods, hyphens, and underscores.
- Username cannot contain spaces.
- Username cannot be longer than XX characters.
- Username must be unique.

Example valid usernames: sam_smith, smith.123

Example invalid usernames: sam smith, _sam

Password Validation Rules: The password validation rules are based on the PasswordEvaluationTestbed-SU26 application and the password recognizer state diagram.

A valid password must follow these rules:

- Password is required.
- Password must be at least 8 characters long.
- Password must not exceed 64 characters.
- Password must include at least one uppercase letter.
- Password must include at least one lowercase letter.
- Password must include at least one number.
- Password must include at least one special character.
- Password must be validated before it is saved or used.

The password recognizer checks password strength, but the team must also make sure the password is not too long. This is important because the assignment specifically requires the application to detect passwords that exceed a reasonable upper limit.

Example valid password: Summerishere!222

Example invalid password: Pa123

Email Address Validation Rules: A valid email address must follow these rules:

- Email address is required.
- Email address must not exceed XXX characters.
- Email address must contain one @ symbol.
- Email address must include a valid domain.
- Email address cannot contain spaces.

Example valid email address: samsmith@asu.edu

Example invalid email address: sam@

Name Validation Rules: First name and last name fields must follow these rules:

- Name is required.
- Name must not exceed XX characters.
- Name may contain letters.
- Name may contain spaces.
- Name may contain hyphens.
- Name may contain apostrophes.
- Name cannot contain numbers.
- Name cannot contain symbols such as @, #, \$, %, or !

Example valid names: Sam, Noah, Kelly-Ann

Example invalid names: Sam12, Noah@

Roles Validation Rules: The application uses roles to control what users can access.

Valid roles are:

- Admin
- Student
- Instructor
- Staff

The role field must not allow any other value.

One-Time Password Validation Rules: The Admin Home Page includes a button for setting a one-time password.

A one-time password must follow these rules:

- One-time password is required when the admin sets or resets it.
- It must follow the same password requirements as a regular password.
- It must not exceed XX characters.
- It must be temporary.
- The user should be required to change it after first use, if supported by the application.

Connection to Testing: Our team should test:

- Valid username accepted
- Invalid username rejected
- Blank username rejected
- Too-long username rejected
- Valid password accepted
- Password without uppercase letter rejected
- Password without lowercase letter rejected
- Password without number rejected
- Password without special character rejected
- Too-long password rejected
- Valid email accepted
- Invalid email rejected
- Blank required fields rejected
- Invalid role rejected
- Valid role accepted
- Password confirmation mismatch rejected

These tests help prove that the application handles input correctly before saving or processing user data.

Summary: This document identifies the required input validation for TP1. The application must validate all user input before processing it. Required fields cannot be blank, text fields must have maximum lengths, passwords must meet strength requirements, and meaningful error messages must be shown when validation fails.

This input validation helps make the application safer, more reliable, and easier for users to understand.